Bangladesh Army International University of Science and Technology

Department of Computer Science and Engineering

Lab Report

Lab Report No : 7

Lab Report Name : Triggers, Indexing.

Course Title : Database Management System Sessional.

Course Code : CSE 314

Name : Nosratee Jahan Naba

ID : 1118005

Level : 3 **Term** : I **Section** : A **Group** : 1

Date of Submission : 22/12/2025 **Semester** : Fall **Year** : 2025

Key Learnings :

- Learned how **triggers** perform automatic actions when INSERT, UPDATE, or DELETE events occur.
- Learned how **indexing** improves query performance and speeds up data retrieval.
- Understood that indexes improve read operations but may slow down write operations.

Code Implementation with Output:

```
CREATE DATABASE University;
USE University;
CREATE TABLE student(
    s_id INT PRIMARY KEY,
    s_name VARCHAR(100)
);

CREATE TABLE audit_student_table(
    id INT AUTO_INCREMENT PRIMARY KEY,
    s_id INT,
    s_name VARCHAR(100),
    action VARCHAR(100),
    action_time TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

```
DELIMITER //
CREATE TRIGGER student_audit
AFTER INSERT ON student
FOR EACH ROW
BEGIN
    INSERT INTO audit_student_table(s_id, s_name, action)
    VALUES (NEW.s_id, NEW.s_name, 'INSERT');
END //
DELIMITER ;


INSERT INTO student
VALUES
(1118005, 'Nosratee Jahan Naba');

SELECT * FROM audit_student_table;

```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

id	s_id	s_name	action	action_time
1	1118005	Nosratee Jahan Naba	INSERT	2025-12-20 16:02:40
NULL	NULL	NULL	NULL	NULL

```
CREATE TABLE course(
    c_id INT PRIMARY KEY,
    c_name VARCHAR(100)
);

INSERT INTO course
VALUES
(313,'DBMS'),
(311, 'CE');


CREATE TABLE log_table(
    id INT AUTO_INCREMENT PRIMARY KEY,
    c_id INT,
    c_name VARCHAR(100),
    action VARCHAR(100),
    action_time TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

```

```
DELIMITER //
CREATE TRIGGER log_update
AFTER DELETE ON course
FOR EACH ROW
BEGIN
    INSERT INTO log_table (c_id,c_name,action)
    VALUES(old.c_id,old.c_name,'DELETE');
END //
DELIMITER ;

DELETE
FROM course
WHERE c_id = 311;

SELECT * FROM log_table;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

id	c_id	c_name	action	action_time
1	311	CE	DELETE	2025-12-20 17:19:22
NULL	NULL	NULL	NULL	NULL

Limit to 1000 rows

```
ALTER TABLE student
add age INT;

DELIMITER //
CREATE TRIGGER check_student
BEFORE INSERT ON student
FOR EACH ROW
BEGIN
    IF NEW.age < 18 THEN
      SIGNAL SQLSTATE '45000'
      SET MESSAGE_TEXT = 'Age must be greater than 18';
    END IF;
END //
DELIMITER ;

INSERT INTO student
VALUES (101, 'A', 17);

INSERT INTO student
VALUES (101, 'A', 20);
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
155	17:40:12	INSERT INTO student VALUES (101, 'A', 17)	Error Code: 1644. Age must be greater than 18	0.000 sec
156	17:40:46	INSERT INTO student VALUES (101, 'A', 20)	1 row(s) affected	0.015 sec

```sql
create table Students(
    id INT PRIMARY KEY,
    dept VARCHAR(100),
    email VARCHAR(100)
);

INSERT INTO Students VALUES
(1, 'CSE','abc@gmail.com'),
(2, 'EEE', 'xyz@gmail.com');


CREATE INDEX idx_email
ON Students(email);

SELECT * FROM Students
WHERE email = 'xyz@gmail.com';
```

Result Grid | Filter Rows: | Edit: | Export/Import:

id	dept	email
2	EEE	xyz@gmail.com
NULL	NULL	NULL

```sql
CREATE INDEX idx_dept
ON Students(dept);

SELECT * FROM Students
WHERE dept = 'CSE';
```

Result Grid | Filter Rows: | Edit:

id	dept	email
1	CSE	abc@gmail.com
NULL	NULL	NULL